theraCUES™
SIGNALS FOR LIFE

Extended Exome Sequencing – Germline

HG-EXOM-EXT

Applications

Method utilizes enhanced target capture of protein-coding regions along with selected regulatory and clinically relevant non-coding regions to comprehensively profile germline genetic variants. Compared to standard exome sequencing, extended exome expands coverage to include untranslated regions (UTRs), splice sites, and disease-associated regulatory elements, improving variant discovery beyond canonical exons. Sequencing reads are aligned to a reference genome followed by variant calling and annotation.

Pros

- Broader coverage target than standard exome
- Improved detection of regulatory variants
- Higher diagnostic yield for rare diseases
- Suitable for unresolved clinical genetic research

Notes

- Intergenic regions out of capture not sequenced
- Structural variant detection is limited
- Coverage variability may occur in GC-rich regions

Input

Genomic DNA (Germline)

DNA Quantity: ≥ 100 ng - 1 µg recommended

Sample Types

Blood | Tissue | Buccal Swabs

Platform

Illumina platforms | 150 × 2
63 mb ~ 14 GB at 100 X

Process Map

Sample DNA Extraction > DNA QC > Capture + Library Prep > NGS Data Acquisition > Data QC & Preprocessing > Reference Alignment > Variant annotation > Gene Rank by Phenotype > Reporting

Standard Deliverables

Download Sample Report

- Sample and Library Quality Control (QC)
- Summary of Methods and Raw Data Quality Control
- Reference Genome Alignment Summary
- Variant Identification and Summary Statistics
- Zygosity and Coding, Variant Annotation and Variant Effect Prediction
- Pathogenic variants(if detected)
- Rare variants by specific phenotypes if described
- *Note: For trio 37 MB | 33 GB data/ 11GB/Sample